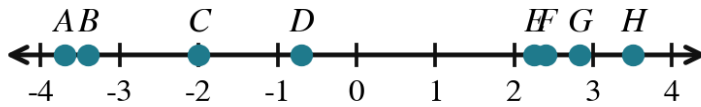


Warm Up:

Name the letter of the corresponding point on the number line.



1. -0.7

→ D

2. 2.4

→ F

3. $\sqrt{8}$

→ G ($\sqrt{8} \approx 2.83$)

4. -2

→ C

5. $2\frac{1}{4}$

→ E

Speak Like A Mathematician:

absolute value: The **DISTANCE** a number is from zero on the number line. Distance is never negative.

Absolute value bars are grouping symbols – simplify everything inside them before taking the absolute value.

Example 1: Evaluating Expressions Involving Absolute Value

Let $n = 4$, $m = -6$, and $p = 2$. Evaluate each expression.

a) $|m|$

→ $|-6| = 6$

b) $-|n|$

→ $-|4| = -4$ – the negative is OUTSIDE the bars, so it survives

c) $|-p|$

→ $|-2| = 2$

d) $-|-m|$

→ $-|-(-6)| = -|6| = -6$

A minus sign inside the bars gets absorbed. A minus sign outside the bars does not.

Example 2: Write an Absolute Value Expression

- The accuracy of a meat thermometer is the positive difference between the temperature reading on the thermometer t and the actual temperature of the meat m . Write two expressions that are both equivalent to the accuracy of the meat thermometer.
→ $|t - m|$ and $|m - t|$ – subtraction order does not matter inside absolute value bars

SC Algebra 1

1.5 – Absolute Value Expressions • **KEY**

- A geologist wants to compare the heights above sea level of a mountaintop in the Adirondacks and one in the Taconic Mountains. If the height of the Adirondack peak a is greater than the height of the Taconic peak t , then which of these statements is NOT equivalent to the difference between the heights of the peaks?

A) $|a - t|$

B) $|t - a|$

C) $|t| - |a|$

D) $|a| - |t|$

→ C. $|t| - |a|$ is negative when $a > t$, so it cannot equal a distance. (A and B are equal; D works only because both heights are positive.)

Example 3: Evaluate the Absolute Value of an Expression

a) Evaluate $|-2xy + 5y|$, if $x = 6$ and $y = -3$.

→ $-2(6)(-3) + 5(-3) = 36 - 15 = 21$; $|21| = 21$

b) Evaluate $|4y - 3xy|$, if $x = 2$ and $y = -2$.

→ $4(-2) - 3(2)(-2) = -8 + 12 = 4$; $|4| = 4$

Example 4: Evaluating Expressions Involving Absolute Value

a) Evaluate $23 - |3 + 4x|$, if $x = 2$.

$$\rightarrow 3 + 4(2) = 11; |11| = 11; 23 - 11 = 12$$

b) Evaluate $2|6x - 20| - 4$, if $x = 3$.

$$\rightarrow 6(3) - 20 = -2; |-2| = 2; 2(2) - 4 = 0$$

c) Evaluate $1.4 - 2.5|5y + 0.6|$, if $y = -3$.

$$\rightarrow 5(-3) + 0.6 = -14.4; |-14.4| = 14.4; 2.5(14.4) = 36; 1.4 - 36 = -34.6$$

Work strictly inside-out: simplify inside the bars, take the absolute value, THEN do the multiplication and subtraction outside.

Tonight's Homework: Paper Practice 1.5 (#1-12)